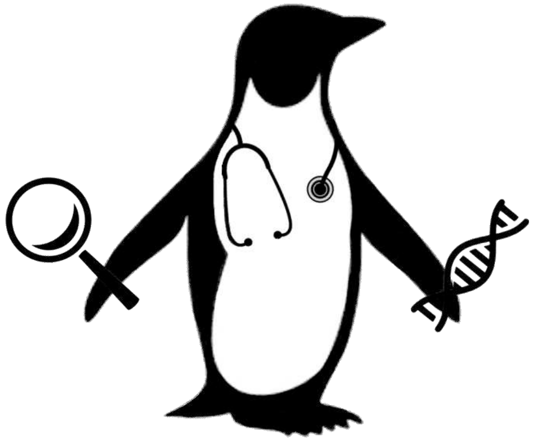
PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice



Elias Crum

European Semantic Web Conference
Ph.D. Symposium (Early Stage)
May 27th, 2024

Outline

Why? — Genomic Data + Semanic Web?

Goals — Reshape how genomic data is stored

How? — Ph.D. plan introduction

Impacts — Deliverables and goals

PENGQUIN

PErsoNal **<u>Genome</u>** QUery IN healthcare and clinical practice

Personal Genome Sequence Data

Human genome data is big…

~3.2 BILLION base pairs ----> ~1 GB

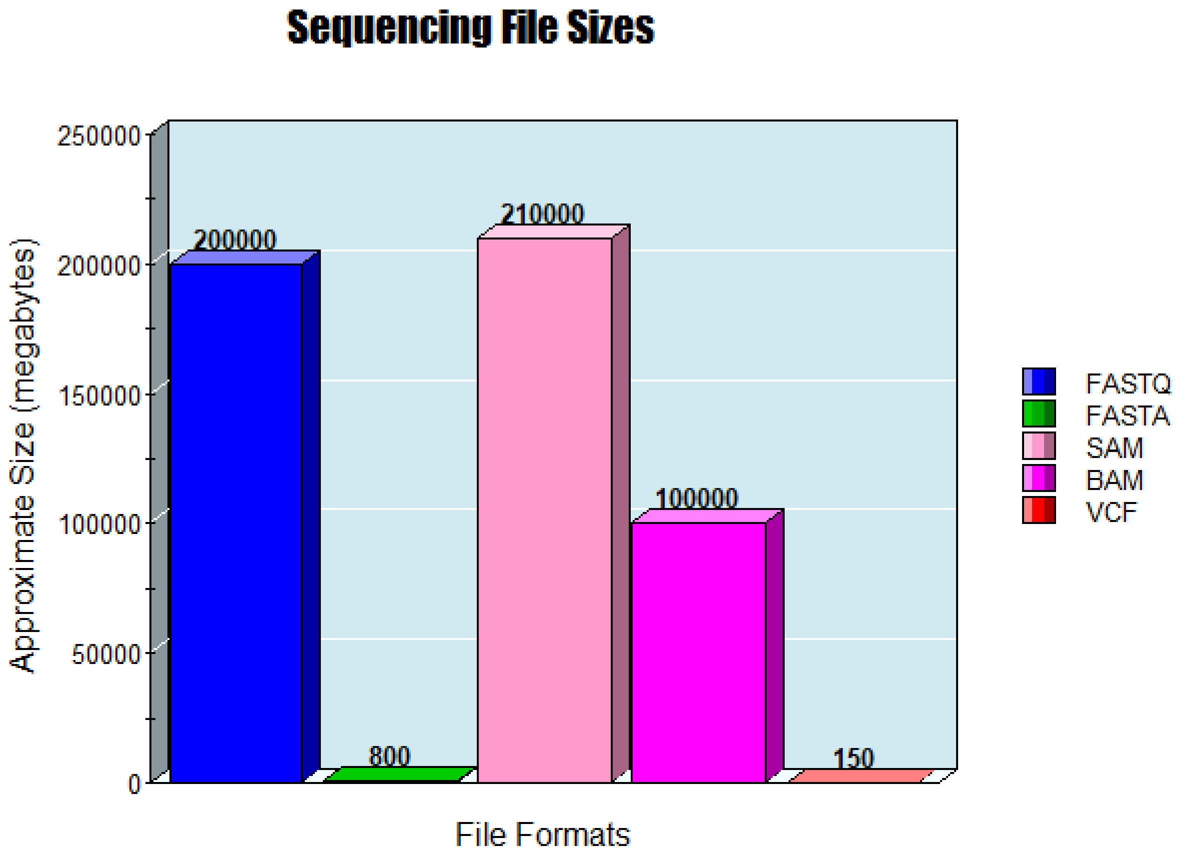
Sequencing output file ----> ~200 GB

Variant Call Format (***VCF***) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

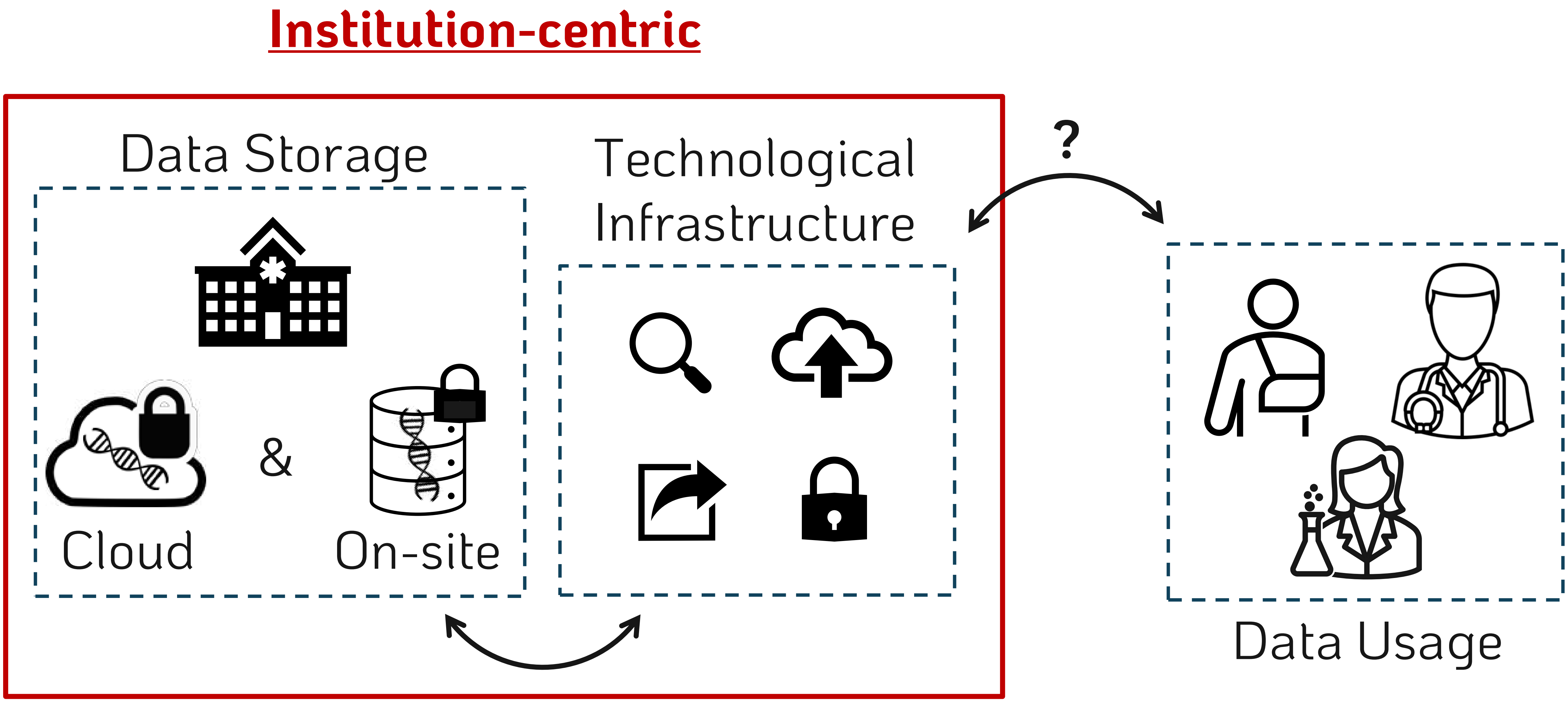
Other considerations

- Storage
- Privacy
- Governance
- Formatting



Bergström et al. SCIENCE 20 Mar 2020.

Semantic Genome Data?



Scaling Genome Data Usage

Research genomic data sharing initiatives:

FHIR

Beacon

OHDSI
OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces → Not much...

Solid Technology

- ➢ Does not solve ALL problems, but offers an environment where improvements are possible
  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

https://solidproject.org/TR/protocol

Research Questions

1. **Can Solid be implemented for genomic data storage?**

   What is the impact on data policy enforcement?

2. **Can (VCF) genomic data be represented as RDF?**

   Can genomic and other medical data be linked semantically?

3. **Can RDF genomic data be queried in a useful way?**

   Can this improve genomic data discoverability for researchers?

Project Goals

Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

Project Goals

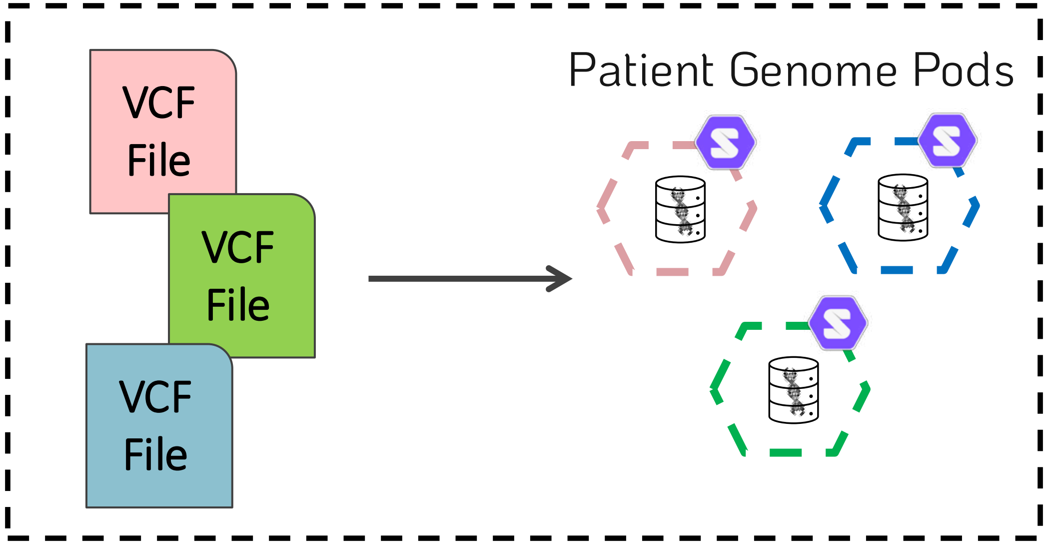
But wait…

This sounds like an **ENGINEERING** project!

Where is the **RESEARCH**?

Workflow Overview

Data Storage



Data Storage

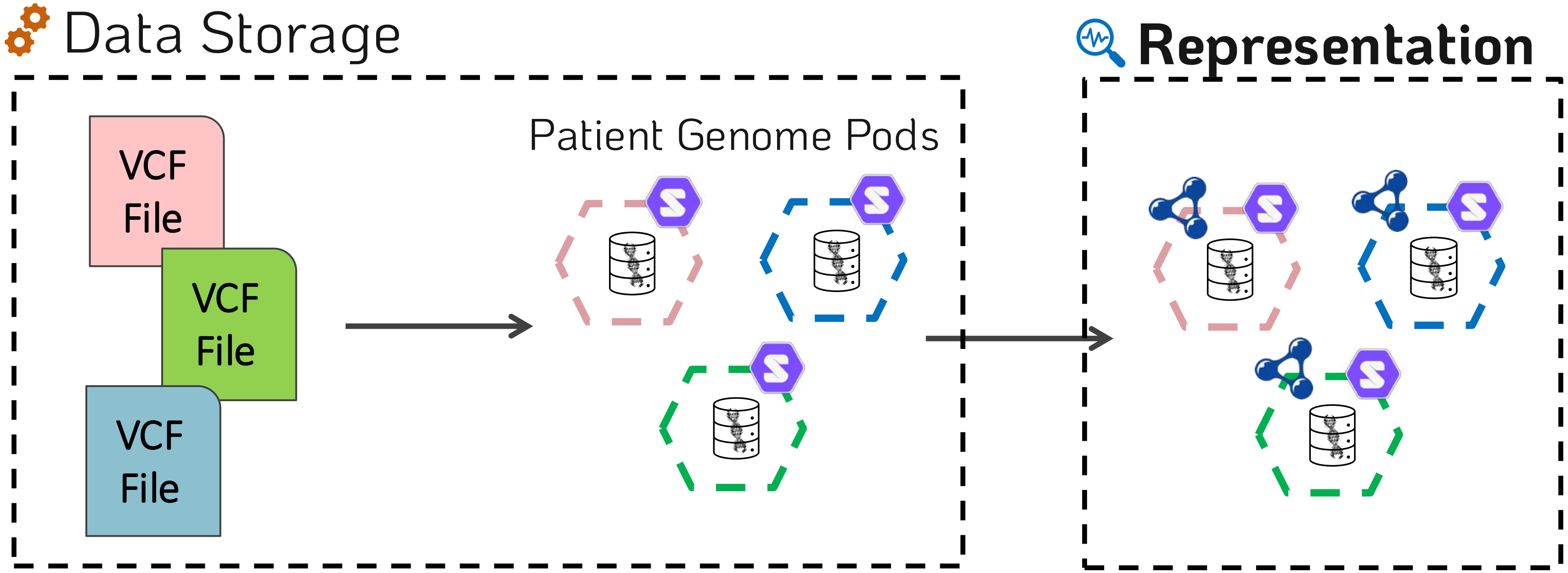
Solid can be implemented for genomic data storage.

Solid pod instances → Community Solid Server[1]

VCF Genomic Data → Illumina Platinum Genomes[2]

[1]https://github.com/CommunitySolidServer/CommunitySolidServer
[2]https://emea.illumina.com/platinumgenomes.html

Workflow Overview

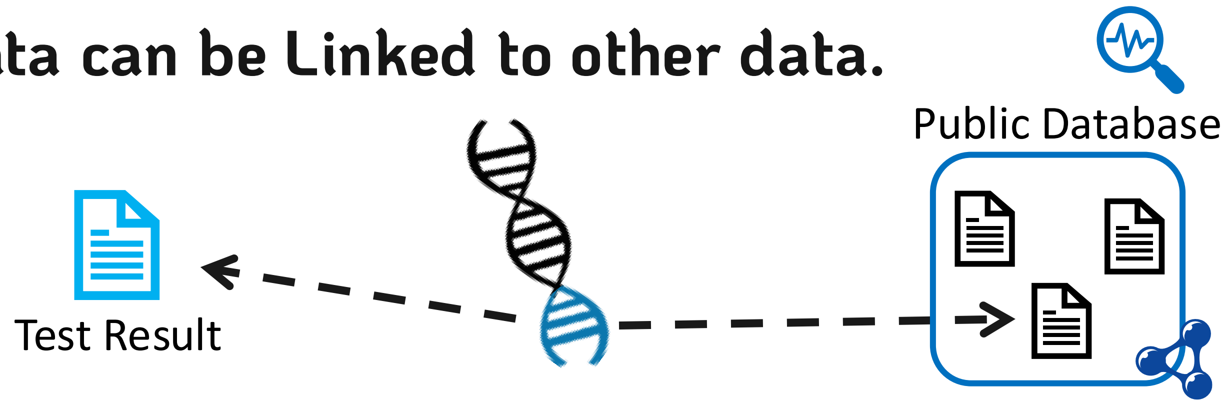


Genome Data in RDF

VCF genomic data can be represented as RDF.

VCF to RDF Vocabularies exist ⟶ SPHN Semantic Interoperability Framework[3] + FHIR HL7 Format[4]

VCF genomic data can be Linked to other data.

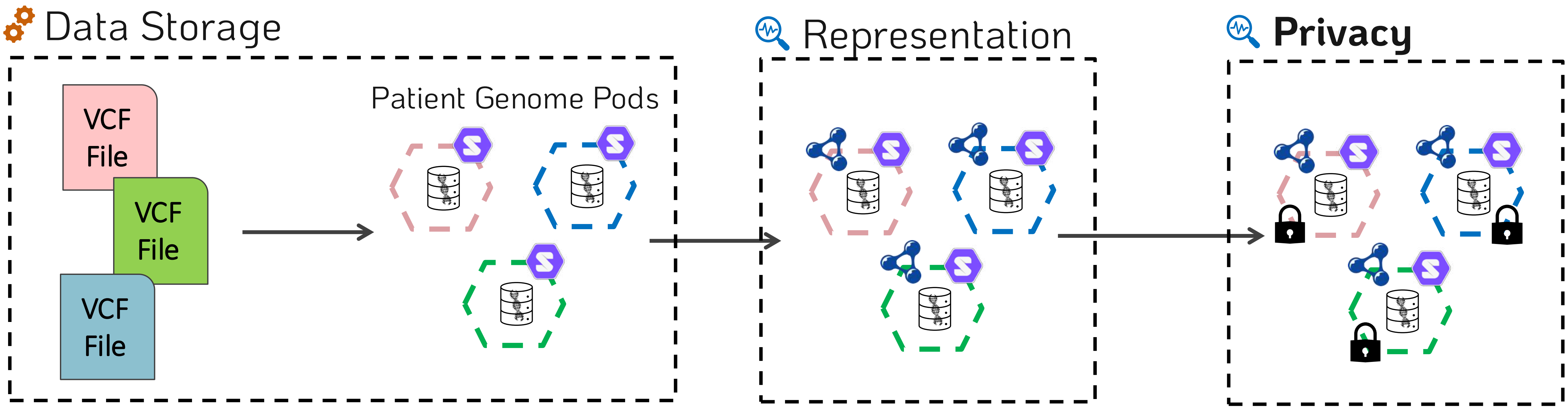


How to automate this?
What vocabularies could be used for the linkages?

[3] van der Horst E, et al. Bridging Clinical and Genomic Knowledge: An Extension of the SPHN RDF Schema for Seamless Integration and FAIRification of Omics Data. 2023.

[4] Dolin RH, et al. vcf2fhir: a utility to convert VCF files into HL7 FHIR format for genomics-EHR integration. 2021.

Workflow Overview



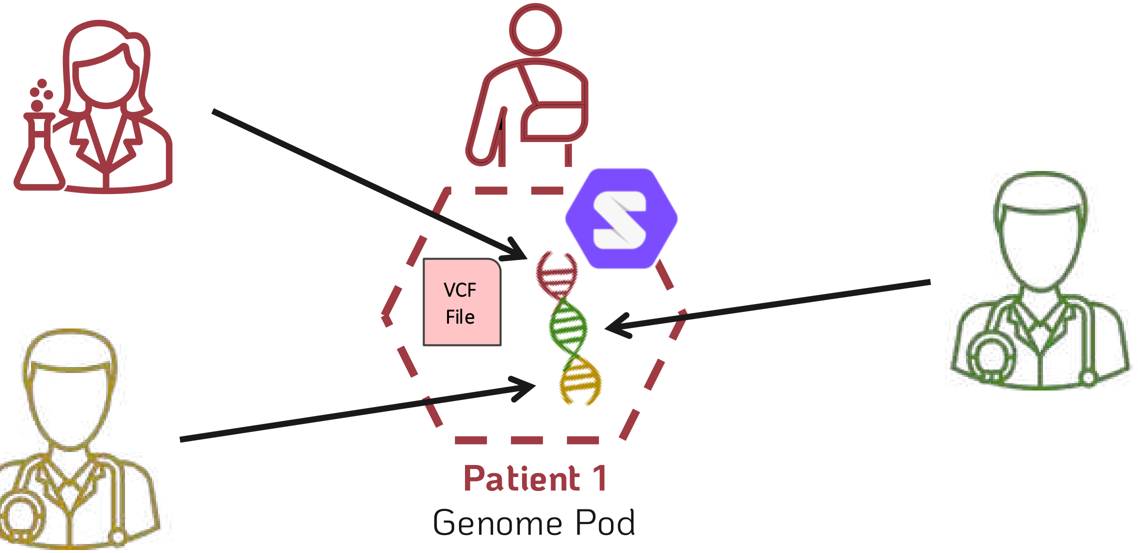
Data Privacy Controls

Granular privacy controls can be implemented in Solid for genomic data.

Web Access Contol → Solid Protocol ACP[5]

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.

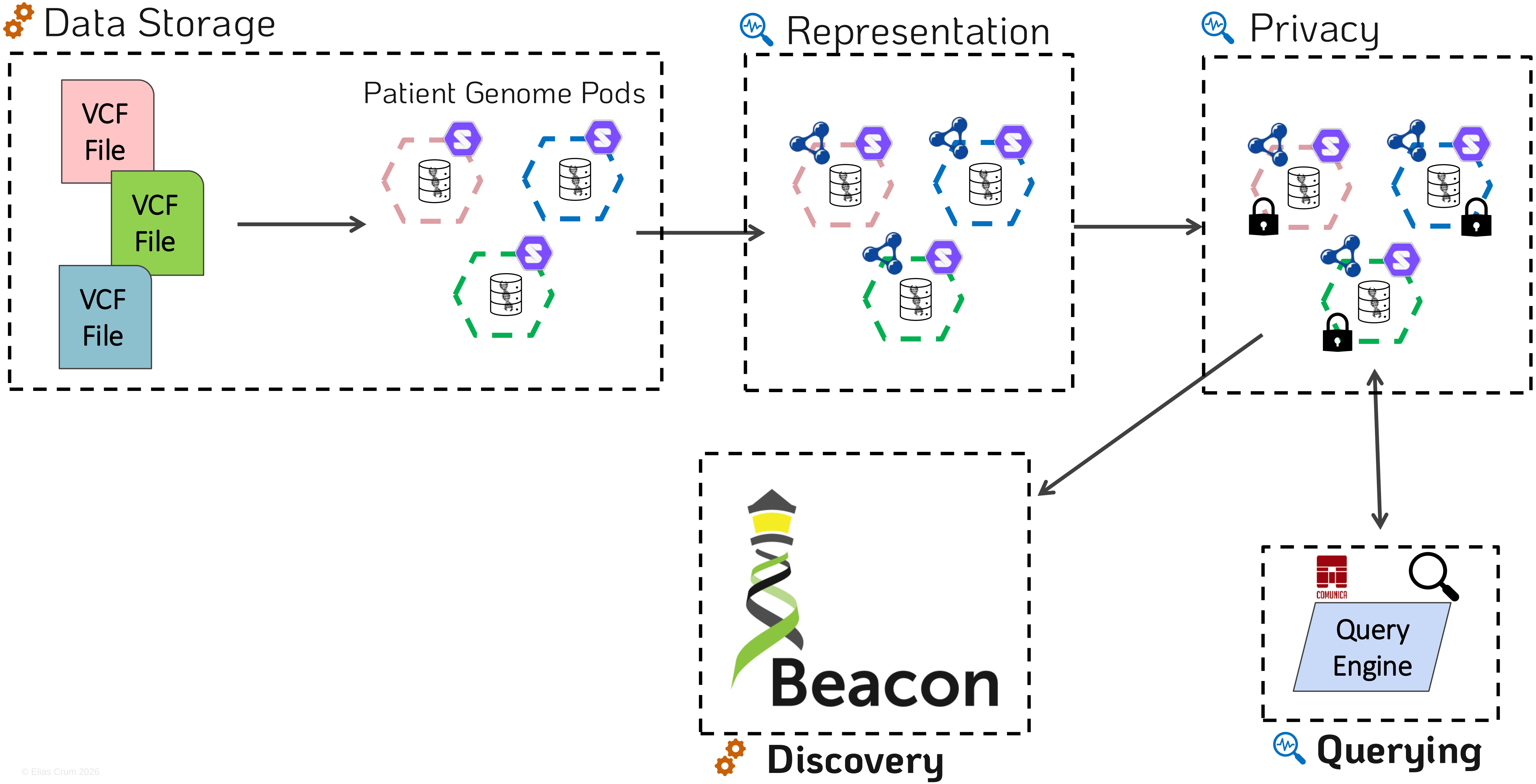


Chromosome is a semantic Class

Mutation is also a semantic Class, above a triple but below a Chromosome

[5] https://solidproject.org/TR/protocol

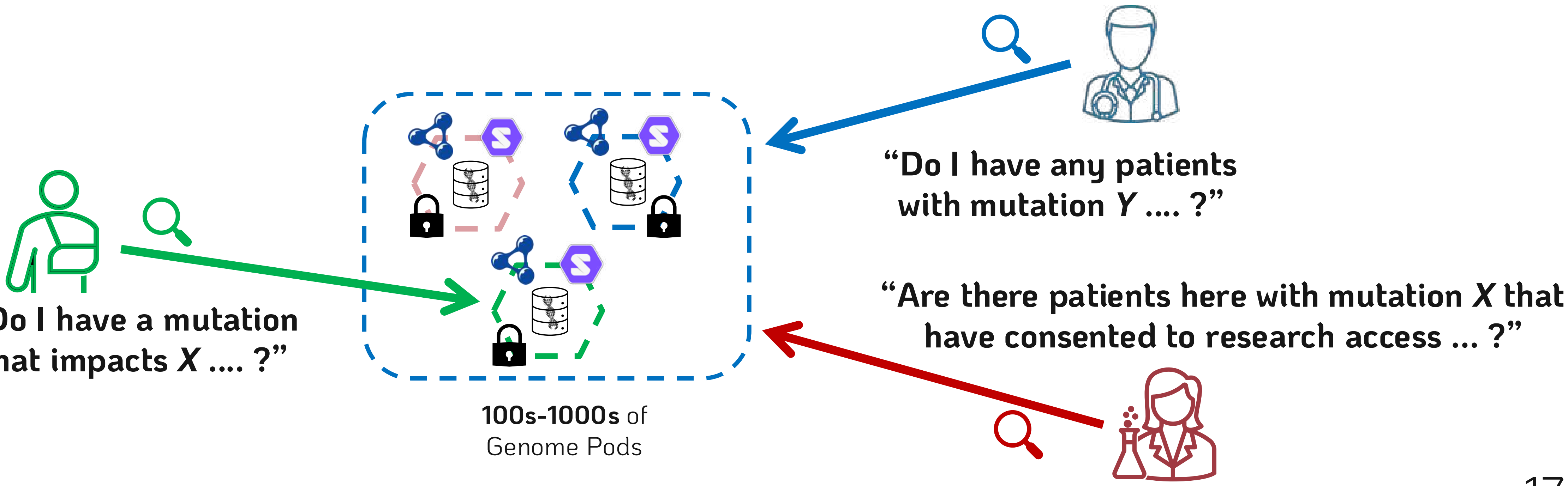
Workflow Overview



Querying Genome Pods

RDF genomic data pods can be queried.

Query Engine → Custom Comunica implementation/engine[6]



[6] https://comunica.dev/

PhD Goals

- ✓ Finished
- Soon
- Future

Basic Functionalities:

- Genomic data storage in Solid pods ✓
- Data privacy management
- Data Sharing via the web ✓

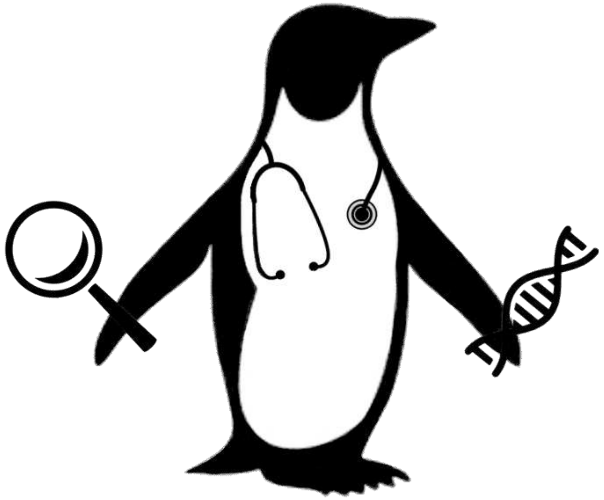
Experimental research:

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?

GHENT UNIVERSITY

IDLab INTERNET & DATA LAB

vito

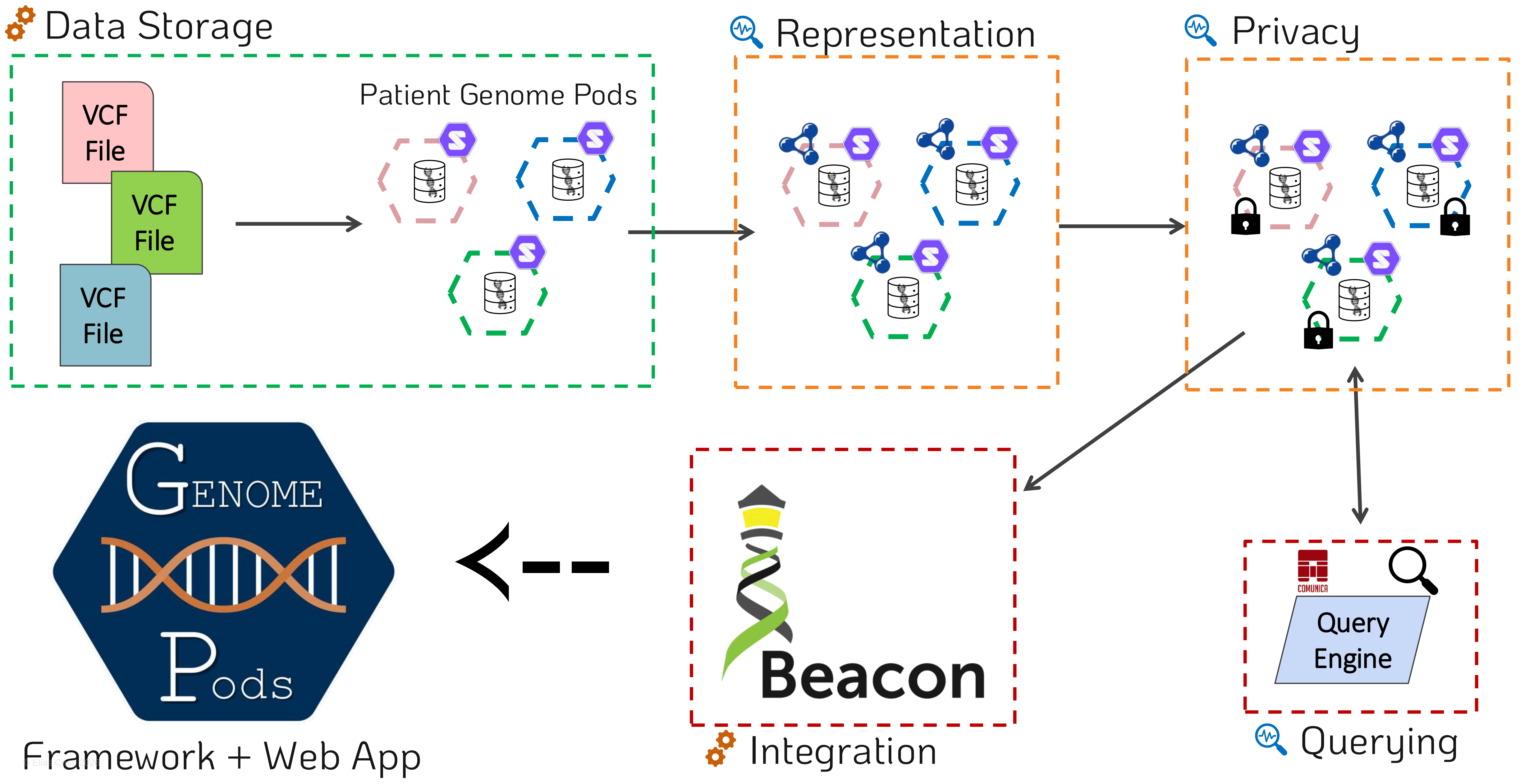


Questions?

Paper

Contact

Workflow Overview



Querying Genome Pods (possibilities)

Single Pod Query

Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed